The burning of nuclear fuel (Box 5.3), in contrast to capturing atoms onto a neutron core (Figure 5.4), can convert only a few tenths of 1 percent of the fuel's mass into heat. In other words, Eddington's heat source (nuclear energy) was roughly 30 times less powerful than Landau's heat source (gravitational energy).[5]

Landau had actually developed a more primitive version of his neutron-core idea in 1931. However, the neutron had not yet been discovered then and atomic nuclei had been an enigma, so the capture of atoms onto the core in his 1931 model had released energy by a totally speculative process, one based on an (incorrect) suspicion that the laws of quantum mechanics might fail in atomic nuclei. Now that the neutron had been known for five years and the properties of atomic nuclei were beginning to be understood, Landau could make his idea much more precise and convincing. By presenting it to the world with a big splash of publicity, he might deflect the heat of Stalin's purge.

In late 1937, Landau wrote a manuscript describing his neutron-core idea; to make sure it got maximum public attention, he took a series of unusual steps: He submitted it for publication, in Russian, to *Doklady Akademii Nauk* (*Reports of the Academy of Sciences of the U.S.S.R*, published in Moscow), and in parallel he mailed an English version to the same famous Western physicist as Chandrasekhar had appealed to, when Eddington attacked him (Chapter 4), Niels Bohr in Copenhagen. (Bohr, as an honorary member of the Academy of Sciences of the D.S.S.R., was more or less acceptable to Soviet authorities even during the Great Terror.) With his manuscript, Landau sent Bohr the following letter:

5 November 1937, Moscow

Dear Mr. Bohr!

I enclose an article about stellar energy, which I have written. If it makes physical sense to you, I ask that you submit it to *Nature.* If it is not too much trouble for you, I would be very glad to learn your opinion of this work.

With deepest thanks.

Yours, L. Landau